GOKULAHEALTH

Testing & QA Plan

Product Management Case Study | V1 → V2

Artifact	Artifact #9 — Testing / QA
Product	GokulaHealth
Scope	V2 Milk-Production Record & Analysis Experience
Testing Method	Functional, usability, data-consistency and offline workflow validation
Status	Test cases prepared — execution pending actual V2 implementation

1. Purpose

This document defines the testing and QA approach for the GokulaHealth V2 product changes. The purpose is to verify that the improved milk-production recording and analysis experience meets the requirements defined in the V2 PRD, user stories and acceptance criteria before release.

2. Testing Objectives

- Verify that morning and evening milk-production records can be entered and saved correctly.
- Verify that monthly production totals and averages are calculated correctly.
- Verify that morning-versus-evening production can be understood from the analytics experience.
- Verify that the redesigned analytics experience is readable and understandable.
- Verify that the core workflow continues to function without internet connectivity.
- Verify consistency between recorded production data and information shown in analytics.
- Verify that empty or insufficient-data states do not present misleading information.

3. Testing Scope

3.1 In Scope

- Milk-production data entry.
- Saving and reviewing production records.
- Monthly total production.
- Monthly average production.
- Morning versus evening comparison.
- Analytics readability and information clarity.
- Offline production-record workflow.
- Data consistency across relevant screens.
- Empty and insufficient-data states.

3.2 Out of Scope

- Features not included in the V2 PRD.
- Production infrastructure or large-scale deployment testing.
- Real-world farmer usability validation beyond the planned product validation activities.
- Performance benchmarking at large production scale unless separately defined.

4. Test Case Design

The test cases below are mapped to the V2 requirements/user stories. They are prepared for execution after the V2 implementation is available. No Pass/Fail results are fabricated in this portfolio artifact.

ID	Requirement / User Story	Scenario	Expected Result	Status	Severity
TC-001	US-01	Enter morning milk yield	Morning entry is saved against the correct cattle/date.	Not Run	High
TC-002	US-02	Enter evening milk yield	Evening entry is saved against the correct cattle/date.	Not Run	High
TC-003	US-03	Save production record	Saved data remains available after leaving and reopening the record.	Not Run	High
TC-004	US-04	Review production records	Saved records are displayed clearly with correct cattle/date/production information.	Not Run	Medium
TC-005	US-05	Verify monthly total	Displayed monthly total matches recorded production values.	Not Run	High
TC-006	US-06	Verify monthly average	Displayed monthly average is calculated correctly and is understandable.	Not Run	High
TC-007	US-07	Morning vs evening comparison	Morning and evening production can be clearly compared.	Not Run	Medium
TC-008	US-08	Analytics readability	Labels, values and charts have sufficient clarity and contrast.	Not Run	High
TC-009	US-09	Offline workflow	Core production-record workflow works without internet.	Not Run	Critical
TC-010	US-10	Data consistency	The same underlying production value is represented consistently.	Not Run	High
TC-011	US-11	Empty/insufficient data	App clearly communicates the data state without misleading calculations.	Not Run	Medium

5. Detailed Test Execution Format

Each test case should be executed using the following evidence fields in the testing spreadsheet:

- Preconditions — what must be true before testing.
- Test Steps — exact actions performed.
- Expected Result — behavior defined by the requirement.
- Actual Result — what actually happened during execution.
- Status — Pass, Fail or Blocked after execution.
- Severity — business/product impact if the test fails.

- Notes — defects, observations or evidence links.

6. Acceptance and Defect Handling

- A test is marked Pass only when the actual result satisfies the expected result.
- A test is marked Fail when the observed behavior does not satisfy the requirement.
- A blocked test should identify the dependency preventing execution.
- High/Critical failures affecting core milk-recording or production-analysis functionality should be resolved before release.
- Usability/readability issues should be documented even when the underlying calculation is technically correct.
- Failed tests should be re-tested after the related defect is fixed.

7. Traceability

The testing spreadsheet provides traceability between the V2 user stories and their corresponding test cases. This connects the PRD → User Stories → Acceptance Criteria → Implementation → Testing flow.

Requirement / User Story	Test Case
US-01 — Morning milk production	TC-001
US-02 — Evening milk production	TC-002
US-03 — Save production record	TC-003
US-04 — Review production records	TC-004
US-05 — Monthly total production	TC-005
US-06 — Monthly average production	TC-006
US-07 — Morning vs evening comparison	TC-007
US-08 — Analytics readability	TC-008
US-09 — Offline workflow	TC-009
US-10 — Data consistency	TC-010
US-11 — Empty/insufficient data	TC-011

8. Agile / Scrum Evidence

Testing is part of the Agile execution cycle rather than a standalone final activity. The V2 backlog is implemented through the Scrum sprint, tested against acceptance criteria, reviewed during the Sprint Review, and improved through feedback and the Sprint Retrospective.

- User Story → Acceptance Criteria → Development → Test Case → Test Result.
- Defects found during testing can return to the backlog for correction.
- Sprint Review can use the tested V2 workflow as product evidence.
- Retrospective can capture what should change in the next iteration.
- Validated feedback can inform the V2.1 backlog.

9. Testing Evidence for Portfolio

- Google Sheets/Excel testing sheet containing the executed test cases.
- Screenshots of important Pass/Fail results.
- Screenshots or links to relevant V2 screens.
- Defect evidence where applicable.
- Final testing summary before release.
- Traceability between requirements, test cases and results.

10. Current Status

The test cases have been prepared as a portfolio-ready QA artifact. Execution is intentionally pending until the V2 implementation is available. Actual Result and Pass/Fail values must be filled only after performing the tests.

11. Next Step

1. Implement the prioritized V2 scope.
2. Execute the prepared test cases against the V2 implementation.
3. Record Actual Result and Pass/Fail status.
4. Fix and re-test failed cases.
5. Prepare V2 Release Notes.
6. Collect post-release feedback and make the V2.1 decision.

Document Control

Document	GokulaHealth V2 Testing & QA Plan
Artifact	Artifact #9
Owner	Bhagyashree Reddy
Status	Prepared for V2 execution
Originality	Portfolio artifact based on the GokulaHealth V1 → V2 product case study